

MetabolismGraph: recovering reaction kinetics and homeostatic regulation from metabolite dynamics with graph neural networks

Abstract

A metabolic network is a bipartite graph in which metabolites are coupled through reactions, each with its own rate constant and stoichiometry. We pose the *inverse* problem: given only the time series of metabolite concentrations and the bipartite graph structure, recover the components that generated the dynamics — the per-reaction rate constants k_j , the substrate-to-rate kinetic law, and the per-metabolite homeostatic regulation. We replace the unknown functions of the forward model with learnable MLPs that operate on the metabolite–reaction graph and a vector of learnable rate constants, and train the graph neural network (GNN) end-to-end to predict dc/dt . On synthetic mass-action systems with known ground truth we study rate-constant recovery and the identifiability obstacles that arise from scale ambiguity, functional compensation, and homeostasis absorption. We outline the path from synthetic data to real metabolomics — starting from fully parameterised kinetic models from BioModels (yeast glycolysis), constraint-based genome-scale models (yeast-GEM), and high-frequency real-time metabolomics during nutrient transitions. This is a companion methods note to our connectome-dynamics studies, which apply the same sign-/structure-locked GNN inverse-modelling recipe to neural circuits.

1 Introduction

Cellular metabolism is a large, sparse reaction network: hundreds of metabolites are interconverted by hundreds of enzyme-catalysed reactions, each proceeding at a rate set by an intrinsic rate constant and the instantaneous concentrations of its substrates. The *forward* problem — predicting concentration dynamics given the network, the kinetics and the rate constants — is classical biochemistry. The *inverse* problem is harder and is our subject: given measured concentration trajectories and the graph of which metabolites enter which reactions, recover the rate constants and the functional form of the kinetics. This is the metabolic analogue of the connectome-to-dynamics inverse problems we study elsewhere, and it shares their central difficulty: the mapping from parameters to dynamics is nonlinear, high-dimensional, and degenerate — many parameter combinations produce nearly identical trajectories.

2 Metabolic networks as bipartite graphs

A metabolic network is naturally a **bipartite graph** with two node types: metabolites and reactions. Edges connect only nodes of different type, and each edge carries an integer *stoichiometric coefficient*. A single-partite metabolite–metabolite graph cannot represent the system, because a reaction couples *several* substrates and products at once through a shared rate constant: collapsing it to pairwise metabolite edges would lose the information that those metabolites participate in the same reaction.

Stoichiometric matrix. The connectivity is summarised by the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{n \times m}$ (n metabolites, m reactions), where S_{ij} is the signed coefficient with which metabolite i enters reaction j : negative for substrates (consumed), positive for products (produced). $\mathbf{S}$ is sparse (each reaction touches only a handful of metabolites), integer-valued (typically in $\{-2, -1, 0, +1, +2\}$), and for mass-balanced reactions its columns sum to zero.

3 Forward model: reaction kinetics and homeostasis

Mass-action kinetics. Under the law of mass action a reaction proceeds faster when its substrates are more abundant, with each substrate raised to the power of its stoichiometric coefficient. For reaction j with rate constant k_j ,

$$v_j = k_j \prod_{k \in \text{sub}(j)} c_k^{|S_{kj}|}, \quad (1)$$

so a reaction consuming two molecules of a substrate depends quadratically on its concentration, and one consuming a single molecule depends linearly. The product over substrates (*multiplicative* aggregation) encodes the requirement that *all* substrates be present simultaneously, and the resulting nonlinear coupling is what allows autocatalytic cycles to sustain oscillations. A simpler *additive* aggregation $v_j = k_j \sum_k c_k^{|S_{kj}|}$ treats substrates as contributing independently and is a useful surrogate for enzyme-mediated reactions that are not strictly mass-action.

Full dynamics. Combining reaction fluxes through $\mathbf{S}$ with a per-metabolite homeostatic restoring force gives the complete forward model

$$\frac{dc_i}{dt} = \underbrace{-\lambda_i (c_i - c_i^{\text{baseline}})}_{\text{homeostasis}} + \underbrace{\sum_{j=1}^m S_{ij} v_j}_{\text{reaction dynamics}}, \quad (2)$$

with v_j from Eq. (1). The homeostatic term pulls each metabolite toward a baseline level c_i^{baseline} at a metabolite-specific rate λ_i ; crucially it is *small* relative to the reaction fluxes, which is what makes it hard to recover (Sect. 5). Eq. (2) spans a family of regimes used for synthetic data: pure reaction ($\lambda = 0$, additive), homeostatic ($\lambda > 0$, additive, convergent), and oscillatory (autocatalytic 3-cycles, multiplicative, sustained oscillations).

4 Inverse problem: the GNN parameterisation

Given observed trajectories $\{c_i(t)\}$ and the bipartite graph, we recover the generating model by replacing every unknown function in Eqs. (1) and (2) with a learnable component on the graph:

$$\frac{dc_i}{dt} = \underbrace{\text{MLP}_{\text{node}}(c_i, a_i)}_{\approx -\lambda_i (c_i - c_i^{\text{baseline}})} + \sum_{j=1}^m S_{ij} k_j \underbrace{\prod_{k \in \text{sub}(j)} \text{MLP}_{\text{sub}}(c_k, |S_{kj}|)}_{\approx k_j c_k^{|S_{kj}|}}. \quad (3)$$

The GNN operates on the metabolite–reaction graph by message passing: substrate states are mapped through MLP_{sub} , aggregated per reaction (multiplicatively for mass-action), scaled by the learnable rate constant k_j , and distributed back to metabolites through $\mathbf{S}$; the node update MLP_{node} adds the

self-regulation term. Training minimises the prediction error on dc/dt end-to-end. The learnable parameters are listed in Tab. 1.

Parameter	Type	Role
$a_i \in \mathbb{R}^d$	embedding vector	per-metabolite identity
k_j	scalar (m of them)	per-reaction rate constant (learned in log-space)
$\text{MLP}_{\text{sub}}(c_k, S_{kj})$	MLP	discovers the kinetic law $c_k^{ S_{kj} }$
$\text{MLP}_{\text{node}}(c_i, a_i)$	MLP	discovers homeostasis $-\lambda_i(c_i - c_i^{\text{baseline}})$

Table 1: Learnable components of the GNN inverse model (Eq. (3)). The stoichiometric matrix $\mathbf{S}$ is either frozen from ground truth (“S given” mode, the focus here) or made learnable (“S learning” mode).

Function discovery. The GNN is not told the kinetic law. MLP_{sub} receives a substrate concentration and its stoichiometric coefficient and must learn the power-law response $c_k^{|S_{kj}|}$ from data alone — in particular it must distinguish linear (c^1) from quadratic (c^2) dependence as the coefficient changes. Likewise MLP_{node} must recover the subtle, small-magnitude homeostatic signal without absorbing dynamics that belong to the reaction terms.

5 Identifiability

The three learnable components interact and can compensate for one another, producing degenerate solutions with high prediction accuracy but poor parameter recovery:

- **Scale ambiguity.** The product $k_j \text{MLP}_{\text{sub}}$ is invariant under $k \rightarrow \alpha k$, $\text{MLP}_{\text{sub}} \rightarrow \text{MLP}_{\text{sub}}/\alpha$. Without anchoring, the MLP absorbs a global scale and shifts every k_j . A centring regulariser pins $\text{mean}(\log k)$ to the known range centre.
- **Functional compensation.** If MLP_{sub} settles on a wrong exponent (*e.g.* $c^{1.5}$ instead of c^2), the rate constants can partly compensate by re-scaling, masking the error.
- **Homeostasis absorption.** MLP_{node} can grow and soak up dynamics that should be explained by reactions, biasing the recovered rate constants.

These are the metabolic counterparts of the sign/scale degeneracies that motivate the structure-locking and regularisation priors in our connectome-dynamics models.

6 Status: one validated working point

It is worth stating plainly where the method stands, because the gap between the synthetic proof-of-concept and real data is large and mostly unbridged. To date there is exactly *one* validated, leak-free working point: recovery of the 256 rate constants k_j on the synthetic oscillatory regime with the stoichiometry $\mathbf{S}$ given. The best single run reaches $R^2 \approx 0.87$ in $\log_{10} k$ (slope ≈ 1 , $\sim 10/256$ outliers), but a cross-seed reproduction puts the honest working point lower, at $R^2 = 0.75 \pm 0.04$ (Sect. 7). This is a genuine result — the loss reads only the concentration dynamics, never any ground-truth label.

Two caveats bound even this point. First, an earlier and more favourable “homeostasis recovery” result was *discarded*: it relied on a supervised contrastive term that read ground-truth metabolite-type

labels during training (label leakage), and once that term was removed the homeostatic function MLP_{node} no longer separates — it stays flat, because single-step dc/dt training cannot see a force that only acts over many steps. Recovering homeostasis honestly (a two-phase / residual scheme, Eq. (2)) is unsolved. Second, the k_j recovery itself carries a run-to-run variance of $\pm 0.2\text{--}0.3 R^2$ at fixed hyperparameters, so the working point is a *distribution*, not a fixed number, and reducing that variance is open. Everything beyond rate-constant recovery on synthetic data — homeostasis, stoichiometry learning, and every real-data rung below — remains to be done.

7 Reproduction and cross-seed robustness

The single validated point (Sect. 6) was reproduced from scratch on a fresh data generation, then cross-validated across training seeds to ask how much of the headline $R^2 \approx 0.87$ is reproducible versus a favourable draw. All runs use the identical leak-free configuration (**k_recovery_winner**: rank-50 oscillatory, **S** given, **aggr** = mul, k -floor and MLP_{sub} -norm priors), varying only the optimisation seed. Tab. 2 reports the recovered rate-constant R^2 .

Seed	training completed	k recovery R^2
77 (published winner)	100%	0.742
42	100%	0.705
79	90%	0.808
7	90%	0.738
<i>mean $\pm$ s.d. (4 mature seeds)</i>		0.748 $\pm$ 0.037
123	30% (stopped)	0.722 (rising)

Table 2: Cross-seed reproduction of the leak-free k_j -recovery working point. Four seeds run to (near-)completion cluster tightly at $R^2 = 0.75 \pm 0.04$; seed 123 was stopped early (to free GPUs for the real-data runs) at 0.72 and still climbing. The runs were deliberately halted once the cross-seed picture was clear.

Read. The working point reproduces *robustly* — the cross-seed spread is small (± 0.04), not the ± 0.2 that single-run exploration logs had suggested — but it reproduces at $R^2 \approx 0.75$, not 0.87. The published 0.87 was the best of many draws; the honest, reproducible recovery of the 256 rate constants is 0.75 ± 0.04 .

Fig. 1 unpacks what that 0.75 means. The seed-77 run recovers the rate constants with *raw* $R^2 = 0.74$ over all 256 reactions, but a *trimmed* $R^2 = 0.98$ once the small outlier tail is removed: 245/256 reactions sit on the identity line at unit slope, and only 11 (4.3%) are mis-estimated by more than 0.3 dex. The raw R^2 is therefore dragged down by a few hard reactions, not by a diffuse failure. We adopt a hard quality gate — the outlier fraction must not exceed 10% — which this run passes comfortably (4.3%). This raw-versus-trimmed split is the number to carry into the real-data comparisons.

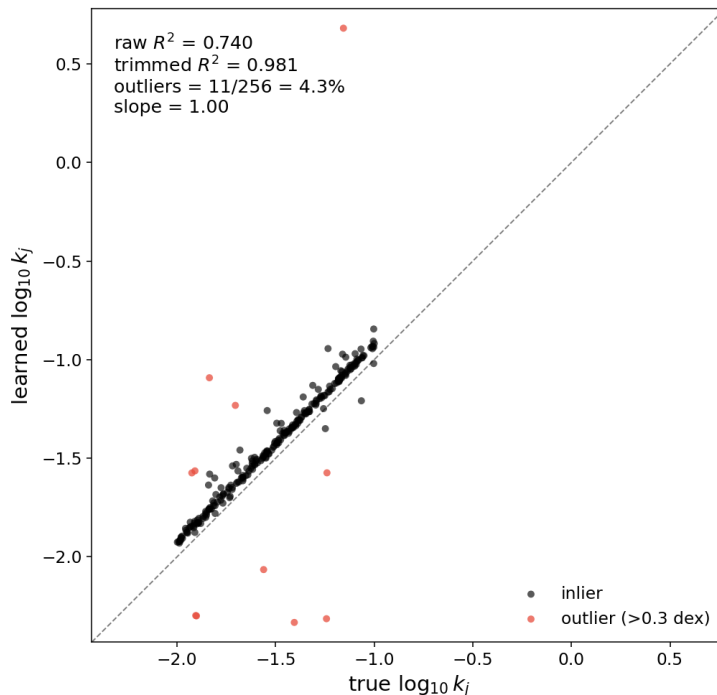


Figure 1: **Toy-model rate-constant recovery (leak-free, seed 77).** Learned vs. true $\log_{10} k_j$ for all 256 reactions after the MLP_{sub} scalar (α) correction; dashed line is identity. Inliers (black) lie on the diagonal at slope 1.00; the 11 red points (4.3%) exceeded the 0.3-dex outlier threshold. Raw $R^2 = 0.74$ over all reactions, trimmed $R^2 = 0.98$ excluding outliers, within the 10%-outlier gate. Reproduced by `figures/k_recovery.py`.

Recovery R^2 vs. rollout R^2 — two different questions. The recovery R^2 above asks whether the *rate constants* are right; it says nothing about whether the model reproduces the *trajectory* when run forward. These can disagree sharply. Fig. 2 free-runs the toy model autoregressively from frame 0 ($\hat{c}(t+1) = \hat{c}(t) + \Delta t f_{\theta}(\hat{c}(t))$): although the rate constants are well recovered (Fig. 1), the rollout *diverges* — rollout $R^2 \ll 0$, Pearson 0.16. The cause is that this working point is trained single-step ($t \rightarrow t+1$) and the homeostatic restoring force (MLP_{node}) is never learned, so integration error compounds with no damping. So for the toy: *recovery* $R^2 = 0.75$ (parameters good) while *rollout* $R^2 \ll 0$ (dynamics unstable) — the two metrics measure different things, and a model can pass one while failing the other. (This is exactly what motivates the autoregressive curriculum of Rung 1, which trains for stable rollout; Fig. 3 shows it reaches rollout $R^2 = 0.99$ *globally*, the opposite trade-off — good rollout, but degenerate parameters.)

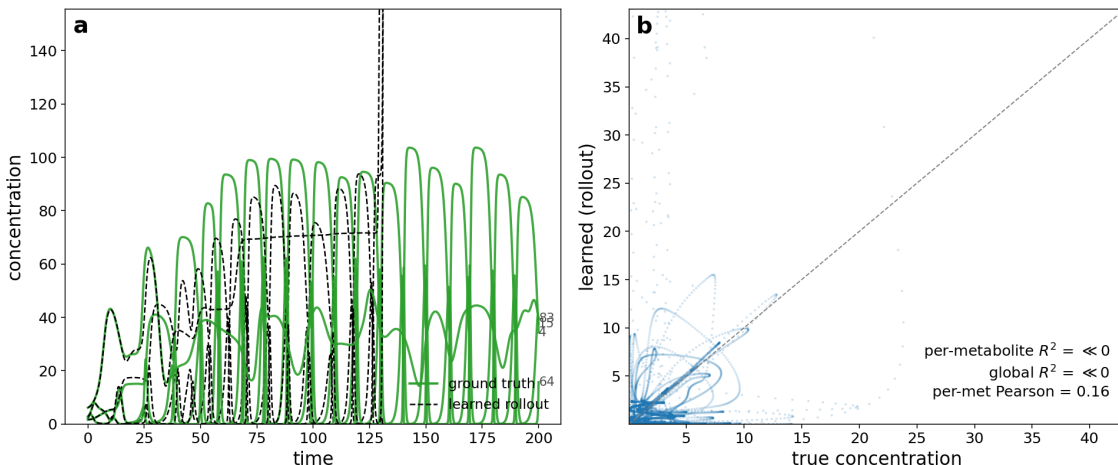


Figure 2: **Toy-model free rollout: good parameters, unstable dynamics.** Autoregressive rollout of the leak-free working point (seed 77) from frame 0. (a) The four highest-variance metabolites: ground truth (green) vs learned free rollout (dashed black), y -axis clipped to $1.5\times$ the true range — the rollout tracks briefly then diverges off-scale. (b) Learned vs true concentration; rollout $R^2 \ll 0$ (the prediction explodes), Pearson 0.16. Despite recovering the rate constants (Fig. 1, $R^2 = 0.74/0.98$), this single-step-trained model does not produce a stable trajectory. Reproduced by `figures/glyco_rollout.py k_recovery_winner`.

8 Toward real metabolomics

The framework is developed and validated on synthetic data where ground truth is known; the goal is real metabolomic time series. Three concrete datasets, now vendored in the repository, stage that transition from fully parameterised kinetics, through genome-scale topology, to a real recording. None of the three is solved yet; this section describes what each provides and what stands between it and a result.

Rung 1 — yeast glycolysis kinetic model (BioModels MODEL1303260010). A small, fully parameterised kinetic model of glycolysis in *S. cerevisiae* (Smallbone et al., 2013), imported from SBML. After separating dynamic from boundary species it gives 20 internal metabolites (+9 fixed boundary species), 30 reactions including iso-enzymes (three hexokinase, three pyruvate-decarboxylase, three glyceraldehyde-3-phosphate variants), with real stoichiometry $\mathbf{S}$ and measured initial concentrations. Crucially the rate laws are *not* mass-action: they mix reversible/irreversible Michaelis–Menten, Hill, and allosteric forms, so $v_j = V_{\max,j} \prod_k [c_k / (K_{m,kj} + c_k)]^{|S_{kj}|}$ rather than a power law. This is the cleanest available ground truth: known $V_{\max}/K_m$ let us score parameter recovery directly. *Status:* the SBML import, $\mathbf{S}$ extraction, and a Michaelis–Menten simulator are in place and generate a 12,000-frame dataset, and we port the connectome-style *autoregressive curriculum* (per-epoch horizon ramp $10 \rightarrow 300$, soft tail-loss, LR co-ramp, gradient clipping) to stabilise the rollout. Two cautions, both found by being sceptical of an initially good number. First, the test rollout looked catastrophic ($R^2 = -390$) only because the evaluator omitted the model’s *boundary stimulus*; supplying it, the curriculum model reproduces the trajectory well (global $R^2 = 0.99$, Fig. 3). Second, that success is misleading: it is *in-sample* (a single trajectory, no held-out run), the per-metabolite R^2 is only 0.15, and the leak-resistant test fails outright — the recovered $V_{\max}$ correlate with ground truth at $R^2 \approx 0$ (slope 0.06, 73% outliers, Fig. 4). The model therefore reproduces the dynamics with

the *wrong* kinetics: a degenerate fit that would not generalise. The curriculum fixed rollout *stability*; the real open problem on real Michaelis–Menten data is parameter *identifiability* — the same scale/compensation degeneracy as the synthetic setup, but here unbroken. (One robust falsification survived both metrics: the config with the lowest *training* loss had the worst rollout *and* the worst recovery, so training loss is not a proxy for either.)

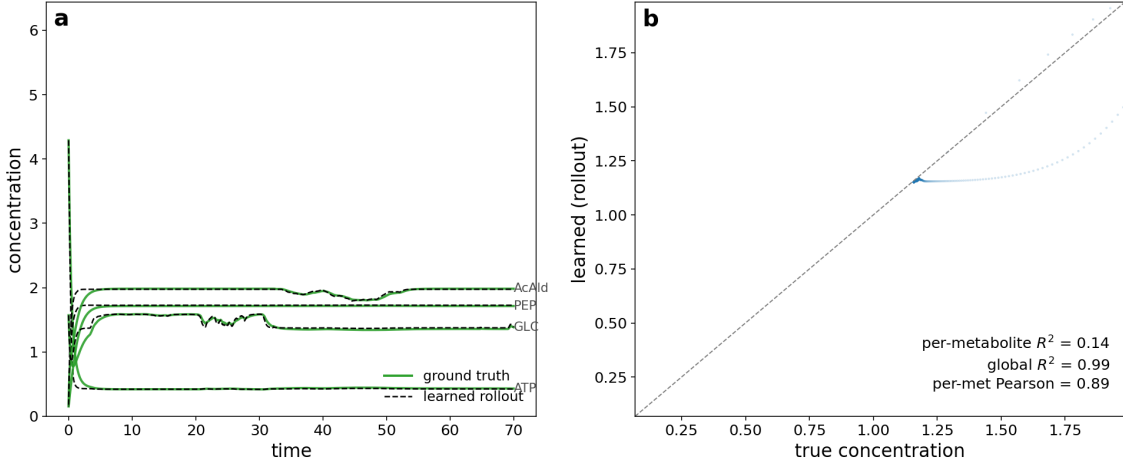


Figure 3: **Rung 1 glycolysis: free-running rollout, ground truth vs learned** (glyco_ar_base, stimulus supplied). **(a)** The four highest-variance metabolites: ground truth (green) vs learned rollout (dashed black). **(b)** Learned vs true concentration over all metabolites/time; global $R^2 = 0.99$ but per-metabolite $R^2 = 0.15$ (the global value is variance-weighted, carried by a few high-variance species). This is *in-sample* (single trajectory, no held-out run). Reproduced by `figures/glyco_rollout.py`.

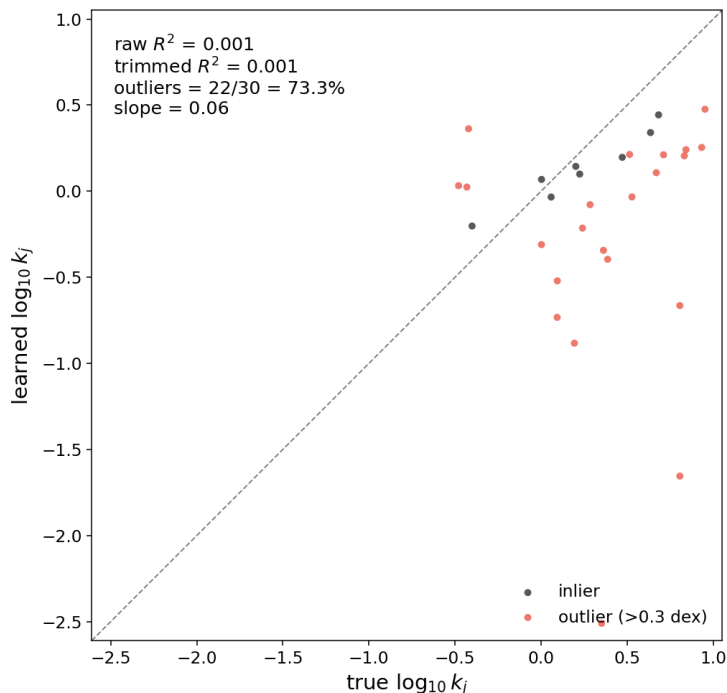


Figure 4: **Rung 1 glycolysis: V_{max} recovery fails** — the leak-resistant test. Learned vs true $\log_{10} V_{max}$ for the 30 reactions: $R^2 \approx 0$, slope 0.06, 22/30 (73%) outliers (fails the 10% gate). Despite the near-perfect in-sample rollout (Fig. 3), the model recovers essentially *none* of the true kinetics — a degenerate fit (right dynamics, wrong mechanism). Reproduced by `figures/k_recovery.py glyco_ar_base`.

Can MLP_{sub} learn the kinetic curvature? A natural suspect for the recovery failures is that the MLP cannot represent the substrate power law $c^{|s|}$. On the synthetic toy this is untestable — that network is 100% $|s|=1$ (linear), so any probe at $|s|\geq 2$ is untrained extrapolation. We therefore built a controlled regime (`mixed_s2`: 31% of substrate edges with $|s|\geq 2$) and compared a plain MLP against the log-space parameterisation $g = \exp(\text{MLP}(\log c, |s|))$ (Fig. 5). Two things hold. First, *with* genuine $|s|\geq 2$ signal the MLP now learns *some* curvature (the toy could not), and log-space learns more: the growth of MLP_{sub} over $c \in [1, 9]$ reaches $24 \rightarrow 32$ for c^2 (true 81) and $28 \rightarrow 57$ for c^3 (true 729), plain $\rightarrow$ log-space. So curvature *does* matter and log-space helps. Second, *both still badly undershoot* the higher powers, and both fail the 10%-outlier gate on this regime (raw k $R^2 = 0.16$ plain vs 0.26 log-space; $\sim 40\%$ outliers). Representing c^3 – c^4 over a wide range remains hard for either parameterisation — a real but partial fix.

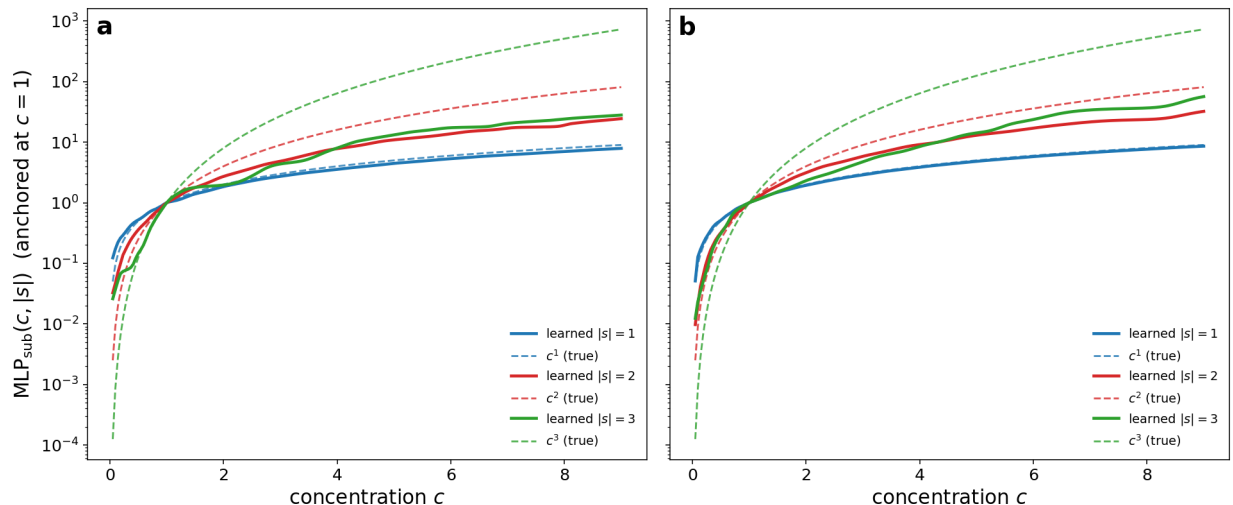


Figure 5: **Curvature recovery on a regime with $|s| \geq 2$ (mixed_s2).** Learned $\text{MLP}_{\text{sub}}(c, |s|)$ (solid) vs the true power law $c^{|s|}$ (dashed) for $|s|=1, 2, 3$, log y -axis, anchored at $c=1$. (a) plain MLP, (b) log-space MLP. Both track c^1 ; both bend upward for c^2, c^3 now that the data contains such reactions, with log-space bending more — but both fall well short of the true c^2 (and far short of c^3). Reproduced by `figures/curvature_compare.py`.

The limit is signal, not capacity. Two further controls localise the residual failure. (i) Training on four diverse trajectories instead of one barely helps (outliers $37.5\% \rightarrow 32\%$), so it is not a data-volume / identifiability-by-replication problem. (ii) A *structured* substrate function $g = c^{a(|s|)}$ — which can represent any exponent *exactly* — still fails (outliers 30.9%): its learned exponent map saturates at $a(s)=1.0, 1.45, 1.64, 1.71, 1.73, 1.75$ for $s=1 \dots 6$, against the true $1, 2, 3, 4, 5, 6$. It nails $a(1)$ (499 reactions) but never reaches the high exponents — because those reactions are *rare* (15/12/8/2 reactions at $|s|=3/4/5/6$). The unrecoverable $\sim 31\%$ are exactly the $|s| \geq 2$ reactions, across all three parameterisations. So the bottleneck for high-order kinetics is neither MLP capacity nor data volume but the *scarcity of high-order reactions*: too little signal to identify a steep exponent. The fix is a structural prior (e.g. integer exponents) or a regime richer in high- $|s|$ reactions — not a better generic function approximator.

Rung 2 — consensus genome-scale model (yeast-GEM). The community-curated genome-scale model of *S. cerevisiae* (Lu et al., 2019): 2,806 metabolites, 4,131 reactions across 14 compartments. It supplies a large, real stoichiometric matrix $\mathbf{S}$ and flux bounds but *no* kinetic rate laws — the true inverse problem. The intended use is to extract a central-carbon subgraph (glycolysis + TCA + fermentation, ~ 100 – 150 reactions), drive it with dynamic-FBA or literature kinetics to generate in-silico dynamics, and fit the GNN in “S given” mode on a realistic topology. *Status*: not started beyond vendoring the model; subgraph extraction and a dynamics generator are still to be written.

Rung 3 — real-time metabolomics during nutrient transitions (real data). High-frequency flow-injection mass spectrometry of live microbial culture (Link et al., 2015), now downloaded as the paper’s supplementary tables. For *E. coli* the released data give a metabolite $\times$ time intensity matrix of 247 annotated ions over 119 evenly spaced time points in three biological replicates (a long batch course), plus a dedicated starvation $\rightarrow$ growth switch series of 302 ions over ~ 148 points;

companion *B. subtilis* and *Y. lipolytica* sets are included. Each ion carries a KEGG annotation (444 distinct KEGG IDs for *E. coli*), though many ions are mass-ambiguous (one m/z maps to several candidate metabolites) and values are *relative* ion intensities, not absolute concentrations. *Status*: the data are in hand but not yet ingested; the work is to deduplicate/annotate ions, build **S** for the measured subset from an *E. coli* genome-scale model (*e.g.* iJ01366) via KEGG→BiGG mapping, and rescale the relative intensities. There is *no* kinetic ground truth here, so success can only be judged by held-out prediction, not by recovered k_j . A parallel mammalian direction is chronic jugular microdialysis in freely moving mice (Nardin et al., 2025) (~ 123 metabolites every 7.5 min over ~ 8 h; 10 components explain 73% of variance).

What transfers and what changes. The core recipe transfers: the bipartite graph, message passing over reactions, function discovery via $\text{MLP}_{\text{sub}}/\text{MLP}_{\text{node}}$, and log-space rate-constant recovery. Real data adds, all at once, the obstacles the synthetic setup omits: non-mass-action (saturating, allosteric) kinetics that MLP_{sub} must learn; a partially known and incomplete **S** (only measured metabolites, ambiguous identities); two orders of magnitude fewer, unevenly spaced time points ($\sim 10^2$ vs $\sim 10^3$); relative rather than absolute concentrations; and external drivers (the nutrient switch) that must enter as a term $f_{\text{ext}}(t)$ in Eq. (2).

What we expect. We do *not* expect the real-data rungs to work soon. The realistic near-term target is Rung 1: get the GNN to recover the published V_{max}/K_m of a real kinetic model under Michaelis–Menten kinetics — a contained problem with ground truth, but one the current code does not yet solve. Rung 2 is a topology-realism check with no kinetic ground truth. Rung 3 (real *E. coli*) is the genuine goal but also where every difficulty above appears together; a first “result” there means only predicting held-out frames of measured dynamics, not recovering mechanism. Tab. 3 summarises the gap. The honest reading: one validated synthetic working point, three real rungs at the data-and-scaffolding stage, and substantial method development (saturating kinetics, honest homeostasis, incomplete-network handling, relative-concentration rescaling) still required before any real-data claim.

Target	Network / kinetics	Ground truth	Status
Synthetic k_j (S given)	random / mass-action	k_j known	works , $R^2 \approx 0.87$
Synthetic homeostasis	random / mass-action	λ_i known	open (single-step blind)
Rung 1: yeast glycolysis	real / MM+Hill	V_{max}, K_m known	rollout in-sample; V_{max} $R^2 \approx 0$ (degenerate)
Rung 2: yeast-GEM subset	real / none (FBA)	none	not started
Rung 3: real <i>E. coli</i>	inferred / unknown	none	data in hand; not ingested

Table 3: Where each target stands. Only the first row is a validated, leak-free result; the rest is the work ahead.

8.1 First look: are the four real datasets amenable?

Before training, we run the same cheap test on *all four* datasets measured by Link et al. (2015) — *E. coli*, *B. subtilis*, *Y. lipolytica*, and a hybrid sample — asking whether each recording carries the structure the GNN needs. The identical four-probe analysis (`figures/amenability.py`) is applied to every organism: SVD low-rank, per-ion lag-1 autocorrelation, example traces, and KEGG coverage against a common reference network (iJ01366; a proxy for core-metabolite mappability — exact for *E. coli*, approximate for the others). For the *Y. lipolytica* fit target the proper yeast network

yeast-GEM raises coverage from 63% to 77% (127/164 ions), confirming the under-estimate from the bacterial proxy. Tab. 4 summarises and Fig. 6 shows the per-organism panels.

What the three probes compute, and how to read them.

- SVD low-rank.** Stack the recording into a matrix $C \in \mathbb{R}^{N \times T}$ (N ions, T time points) and z -score each row over time, so the probe sees temporal *shape* not absolute scale. Its singular values $\sigma_1 \geq \sigma_2 \geq \dots$ give the variance carried by each independent mode, $\sigma_i^2 / \sum_j \sigma_j^2$; rank_{99} is the smallest number of modes whose cumulative variance reaches 99%. *Interpretation:* it counts the effective dimensionality of the dynamics. A small rank_{99} relative to N means the metabolites move in a coordinated, low-dimensional way — shared drivers and strong correlations — which is exactly the structure a compact network model can fit; $\text{rank}_{99} \approx N$ would mean every ion moves independently and no parsimonious mechanism exists. We compare the *fraction* rank_{99}/N against the synthetic regime ($\sim 45\%$).
- Per-ion lag-1 autocorrelation.** For each ion’s mean-subtracted trace $x(t)$, $\rho_1 = \sum_t x(t)x(t-1) / \sum_t x(t)^2$ — the correlation of the signal with itself shifted by one frame; we report the median over ions and the fraction with $\rho_1 > 0.5$. *Interpretation:* it measures temporal smoothness. $\rho_1 \rightarrow 1$ means consecutive samples are nearly equal — a smooth, slowly varying trace whose finite-difference derivative dc/dt is well defined and carries signal; $\rho_1 \rightarrow 0$ means white-noise-like, so the measured dc/dt is dominated by noise. This is the decisive test for our setting, since the GNN learns $dc/dt = f(c)$: a derivative made of noise cannot be fit, however low-rank or well-annotated the data are.
- KEGG coverage.** Each ion is annotated with one or more KEGG compound identifiers from its m/z (an ion matching several formulae is *ambiguous*); we intersect those identifiers with the metabolites present in a reference genome-scale network (iJO1366), and coverage is the number of ions hitting at least one model metabolite. *Interpretation:* it asks whether a stoichiometry $\mathbf{S}$ — the reaction graph the GNN operates on — can be built for the measured species. A metabolite absent from every known reaction has no edges and cannot be modelled mechanistically, so high coverage means the measured set sits inside a real network; the *unambiguous* count (single KEGG per ion) flags how many identities are clean versus mass-degenerate.

Organism	ions $\times$ t	rank ₉₉	autocorr med.	in iJO1366	dc/dt
<i>E. coli</i>	247 $\times$ 119	102 (41%)	0.27	190 (77%)	noisy
<i>B. subtilis</i>	239 $\times$ 119	98 (41%)	0.41	160 (67%)	noisy
<i>Y. lipolytica</i>	164 $\times$ 59	51 (31%)	0.62	103 (63%)	smooth
Hybrid	338 $\times$ 60	55 (16%)	0.24	155 (46%)	noisy

Table 4: Amenability of the four measured datasets. rank_{99} is the number (and fraction) of singular components for 99% variance; autocorr is the median per-ion lag-1 autocorrelation; coverage is the count mapped onto iJO1366. All four are low-rank and reasonably network-mappable; only *Y. lipolytica* has smooth (dc/dt -learnable) dynamics.

Does it generalise? Two of the three signals hold across all four datasets: the dynamics are *low-rank* ($\text{rank}_{99} = 16\text{--}41\%$ of ions) and a large share of metabolites are core/conserved (coverage 46–77%). The third — temporal smoothness of dc/dt — does *not* generalise: *E. coli*, *B. subtilis* and the hybrid are noise-dominated (median lag-1 autocorrelation 0.24–0.41), whereas ***Y. lipolytica*** is

smooth (median 0.62, 59% of ions above 0.5). The actionable consequence: the most promising target for an actual real-data fit is *not E. coli* (best network coverage but noisiest) but *Y. lipolytica* — smooth, learnable dynamics that are still low-rank and network-mappable. The hybrid is least amenable (lowest coverage, noisy), though most strongly low-rank. A real fit should therefore start from *Y. lipolytica*, restricted to its smooth, high-variance, network-mapped ions and judged by held-out prediction — not by mechanism recovery across the whole noisy measurement.

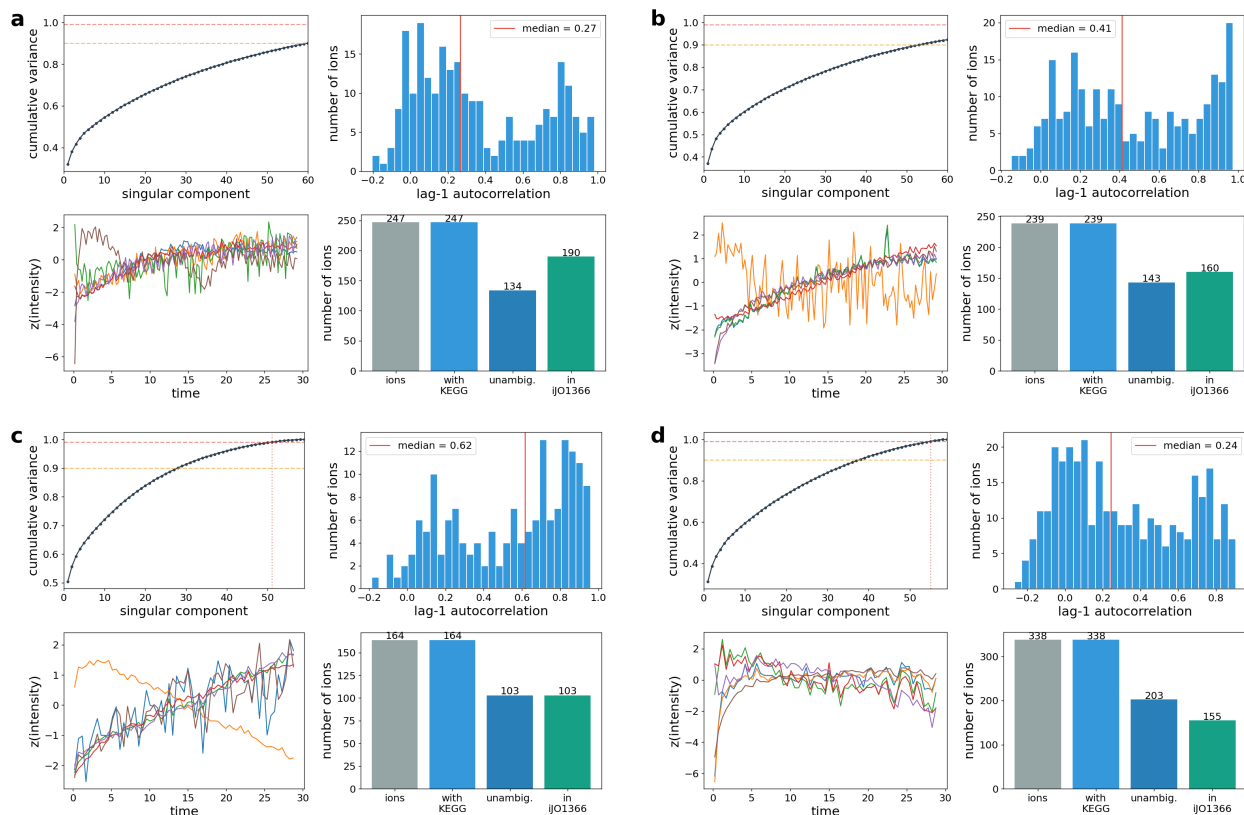


Figure 6: **Identical amenability probes across the four measured datasets:** (a) *E. coli*, (b) *B. subtilis*, (c) *Y. lipolytica*, (d) hybrid. Within each panel, top-left: SVD cumulative variance (low-rank); top-right: per-ion lag-1 autocorrelation; bottom-left: six highest-variance traces (z -scored); bottom-right: KEGG coverage against iJ01366. Low-rank structure and network coverage recur in all four; only *Y. lipolytica* shows smooth, slowly varying traces and a right-shifted autocorrelation distribution. Reproduced by `figures/amenability.py`.

Data and code

Synthetic generators, the GNN model, and the LLM-driven exploration loop are in the `MetabolismGraph` repository; an overview minisite is at <https://allierc.github.io/MetabolismGraph/>. The reference material is vendored under `papers/`: the two yeast models, BioModels MODEL1303260010 (<https://github.com/biomodels/MODEL1303260010>) and yeast-GEM (<https://github.com/SysBioChalmers/yeast-GEM>); and the Link et al. (2015) real-time-metabolomics supplementary tables under `papers/nmeth3584/`. The Rung-1 SBML→dynamics bridge is the `glycolysis_yeast` config/dataset; the leak-free synthetic k_j -recovery working point is the `k_recovery_winner` config.

A Weekend hypothesis ledger (live)

This appendix is updated autonomously every few hours during unattended runs. Each row is a hypothesis with its experiment and a *validated* / **FALSIFIED** / *in progress* verdict. Falsifications are recorded as plainly as positive results.

Date	Hypothesis	Experiment	Verdict
2026-06-05	H1: winner reproduces k $R^2 \approx 0.87$	seed 77, leak-free, S given	0.74 raw / 0.98 tr
2026-06-05	H2: k R^2 robust across seeds	CV 4 seeds, S given	<i>validated</i> (0.75±0
2026-06-05	glycolysis rollout $R^2 = -390$ is real	data_test omits stimulus	FALSIFIED (eval l
2026-06-05	H4: AR curriculum <i>recovers</i> glycolysis	rollout (w/ stim) + $V_{\max}$ recovery	<i>no</i> : rollout in-san
2026-06-05	rollout R^2 = mechanism learned?	global rollout vs $V_{\max}$ recovery	FALSIFIED (deger
2026-06-05	lowest train loss = best rollout?	clip05 (min loss) vs base	FALSIFIED (worst
2026-06-05	H5: real <i>E. coli</i> is low-rank	SVD of 247×119 matrix	<i>validated</i> (rank ₉₉ =
2026-06-05	H6: real dc/dt is learnable/smooth	per-ion lag-1 autocorrelation	FALSIFIED (med
2026-06-05	H7: ions map to a real network	KEGG $\cap$ iJO1366 / e_coli_core	<i>validated</i> (190/24
2026-06-05	generalises to 4 organisms?	same probes: Bsubt/Ylipo/hybrid	low-rank & cover q
2026-06-05	best real fit target	cross-organism dc/dt smoothness	<i>Y. lipolytica</i> (auto
2026-06-05	log-space MLP_{sub} improves toy k	k_logspace vs plain MLP	<i>validated</i> (0.74→0
2026-06-05	toy degeneracy is a c^2 failure	$ s $ distribution of toy network	FALSIFIED (toy a
2026-06-05	log-space MLP_{sub} fixes glyco $V_{\max}$	glyco_logspace, leak-resistant	FALSIFIED ($R^2=0$
2026-06-06	MLP_{sub} learns c^2 given $ s \geq 2$ data?	mixed_s2: plain vs log-space	<i>partly</i> (log > plain
2026-06-06	more trajectories break the $ s \geq 2$ degeneracy?	mixed_s2: 1 vs 4 diverse trajectories	FALSIFIED (37→3
2026-06-06	exact power-law $c^{a(s)}$ fixes high- $ s $?	learned exponent map $a(s)$	FALSIFIED ($a(s)$ -
2026-06-06	right network for the <i>Y. lipolytica</i> fit	KEGG $\cap$ yeast-GEM vs iJO1366	yeast-GEM 77% (
2026-06-06	is the <i>Y. lipolytica</i> fit well-posed?	measured subnetwork in yeast-GEM	<i>yes</i> : $S \sim 401$ met
2026-06-06	a clean (fully-observed) first fit exists?	reactions with all participants measured	<i>yes</i> : 164/740 rxn
2026-06-07	does good k -recovery imply good rollout?	toy free rollout vs recovery	NO (recovery 0.75

Table 5: Live experiment ledger. The single-step rollout on the real Michaelis–Menten glycolysis model diverges ($R^2 = -390$, Pearson 0.19), motivating the autoregressive curriculum (Sect. 6) ported from the connectome models: per-epoch horizon ramp 10→300, soft tail-loss, LR co-ramp, gradient clipping. Implementation is in place and smoke-validated (runs, finite loss); the 5-config sweep is queued behind the CV runs.

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